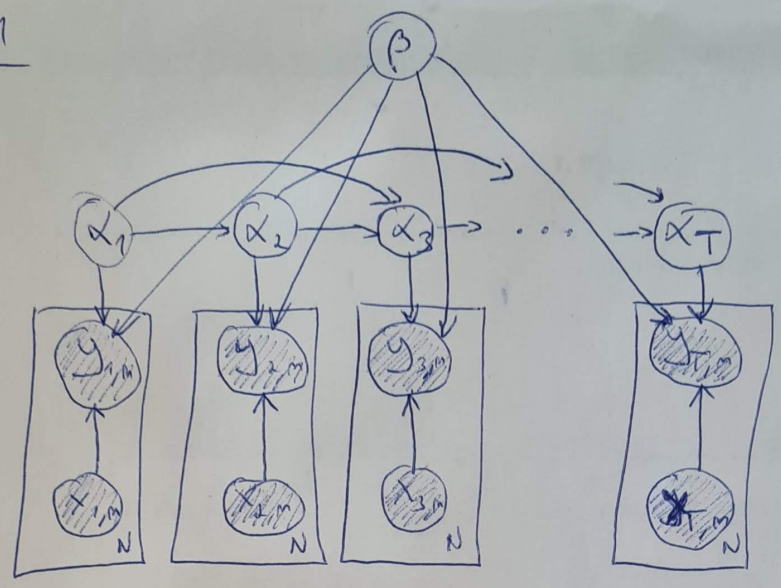
Part 1

1)



all bold

2) $$p(\mathbf{y}, \boldsymbol{\beta}, \boldsymbol{\alpha} \mid \mathbf{X}) = \mathcal{N}(\alpha_1 \mid 0, \sigma_\alpha^2)\, \mathcal{N}(\alpha_2 \mid \alpha_1, \sigma_\alpha^2)\, \mathcal{N}(\beta \mid 0, \sigma_\beta^2 I)$$

$$\prod_{t=3}^{T} \mathcal{N}\big(\alpha_t \mid \underbrace{f(\alpha_{t-1}, \alpha_{t-2})}, \sigma_\alpha^2\big)$$

$$\prod_{m=1}^{N} \text{Bernoulli}\big(y_{t,m} \mid \text{Sigmoid}(\alpha_t + \beta^T \mathbf{x}_{t,m})\big)$$

The question was vague regarding how α_t depends on α_{t-1} and α_{t-2}, so many options will be acceptable here.

Part 2)

1) you just had to realize that, since the posterior is multimodal, you need to break the symmetry somehow through a careful initialization, otherwise the 4 chains will converge to different solutions. this refers to both the initial state and the step size, because if the latter is too big, then it can jump between modes. the concrete method for initializing the state is less important...

2) For the same reasons as above, you want it to be mode-seeking, so $KL(q \| p)$ is the right choice.

Part 3

1) $Bel(S) = ?$

$$\lambda(F) = \pi(F) = \begin{cases} 0, & F=0 \\ 1, & F=1 \end{cases}$$

$$\lambda(H) = \pi(H) = \begin{cases} 0, & H=0 \\ 1, & H=1 \end{cases}$$

$$\pi(S) = \begin{cases} 0.9, & S=0 \\ 0.1, & S=1 \end{cases}$$

$$\pi(R) = \begin{cases} 0.8, & R=0 \\ 0.2, & R=1 \end{cases}$$

$$\lambda_{F,R}(R) = \sum_{F \in \{0,1\}} \lambda(F)\, p(F|R) = p(F=1|R) = \begin{cases} 0.2, & R=0 \\ 1, & R=1 \end{cases}$$

$$\pi_{R,H}(R) = \pi(R)\,\lambda_{F,R}(R) = \begin{cases} 0.8 \times 0.2 = 0.16, & R=0 \\ 0.2 \times 1 = 0.2, & R=1 \end{cases}$$

$$\lambda_{H,S}(S) = \sum_{H\in\{0,1\}} \lambda(H) \sum_{R\in\{0,1\}} p(H \mid R,S)\cdot \pi_{R,H}(R)$$

$$= \sum_{R\in\{0,1\}} p(H=1 \mid R,S)\,\pi_{R,H}(R) = \begin{cases} 0 + 1 \times 0.2 = 0.2, & S=0 \\ 0.9 \times 0.16 + 1 \times 0.2 = 0.34, & S=1 \end{cases}$$

$$\lambda(S) = \lambda_{H,S}(S)$$

$$Bel(S) \propto \lambda(S)\,\pi(S) = \begin{cases} 0.2 \times 0.9 = 0.18, & S=0 \\ 0.34 \times 0.1 = 0.034, & S=1 \end{cases}$$

Normalizing, gives:

$$Bel(S) = \begin{cases} 0.84, & S=0 \\ 0.16, & S=1 \end{cases}$$